Steering Vectors Differ Across Personas

Jacob Davies Rhea Karty Nathaniel Mitrani
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Key Finding: Steering vectors for the same trait encode qualitatively distinct representations depending on the active persona—differing in both magnitude and direction (cosine similarity). Traits heavily optimised during RLHF (such as honesty) show reduced persona diversity, suggesting post-training collapses representational variation.

Motivation

Character traits like sycophancy correspond to **linear directions in activation space**—*persona vectors*—that can monitor and steer model behaviour (Chen et al., 2025). The **Assistant Axis** captures how "assistant-like" a model is (Lu et al., 2026). Both extract vectors from a single baseline persona. But models are deployed across diverse modes—chat assistants, autonomous agents, roleplayed characters—each a distinct persona state.

Open question: Do steering vectors transfer across personas, or do personas fundamentally reshape trait representations?

Method

10 personas via system prompts: Farmer, Politician, Therapist, Drill Sergeant, Street Hustler, Professor, Tech CEO, Kindergarten Teacher, Surgeon, Con Artist.

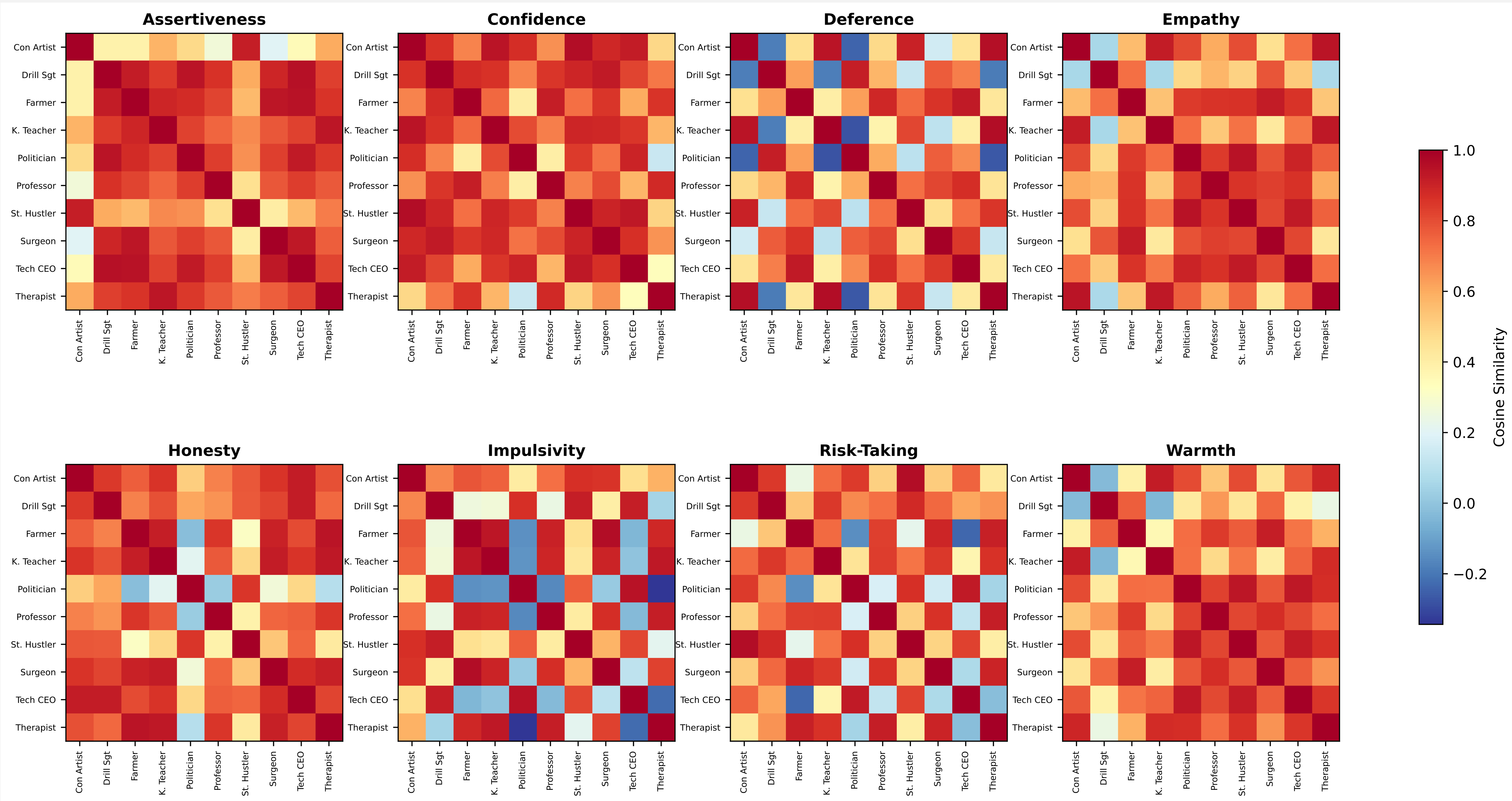
8 traits: assertiveness, empathy, risk-taking, honesty, confidence, deference, warmth, impulsivity.

Contrastive activation addition (CAA):[1] 500 A/B multiple-choice scenarios per trait. Model selects the positive or negative option; activations extracted at the answer token. Contrastive vector from difference in means: $\vec{v}_{\text{trait}} = \text{mean}(\vec{a}_{\text{pos}}) - \text{mean}(\vec{a}_{\text{neg}})$

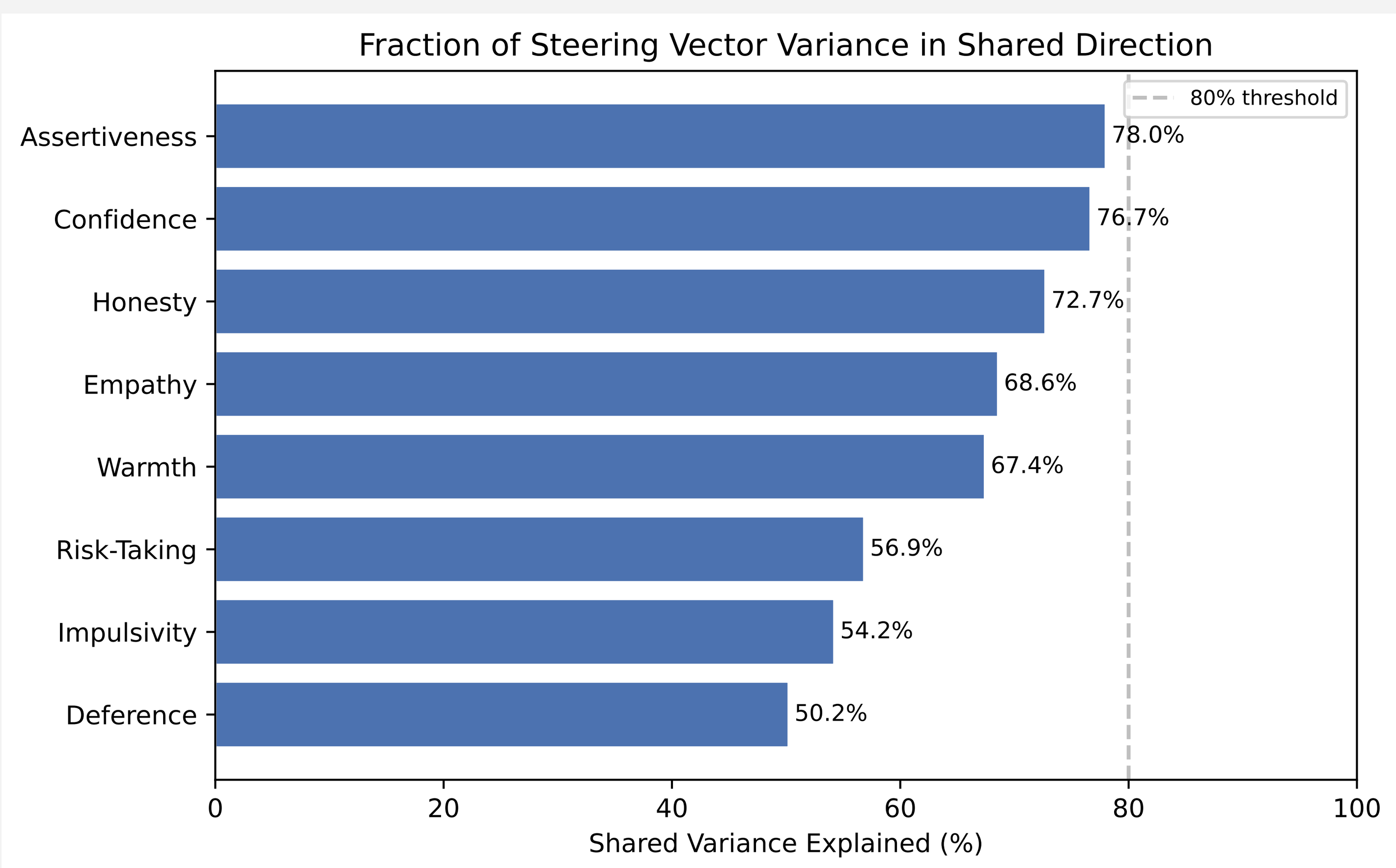
Model: Gemma-2-27B-IT, layer 22/46. Comparison via **cosine similarity** and **behavioural evaluation** (Claude LLM judge).

Geometric Analysis: Cross-Persona Cosine Similarity by Trait

Each heatmap shows pairwise cosine similarity between persona-specific steering vectors for one trait. Red = similar encoding; blue = distinct. **Honesty**, **Assertiveness**, and **Confidence** are highly shared; **Risk-Taking**, **Impulsivity** and **Deference** are more persona-specific.

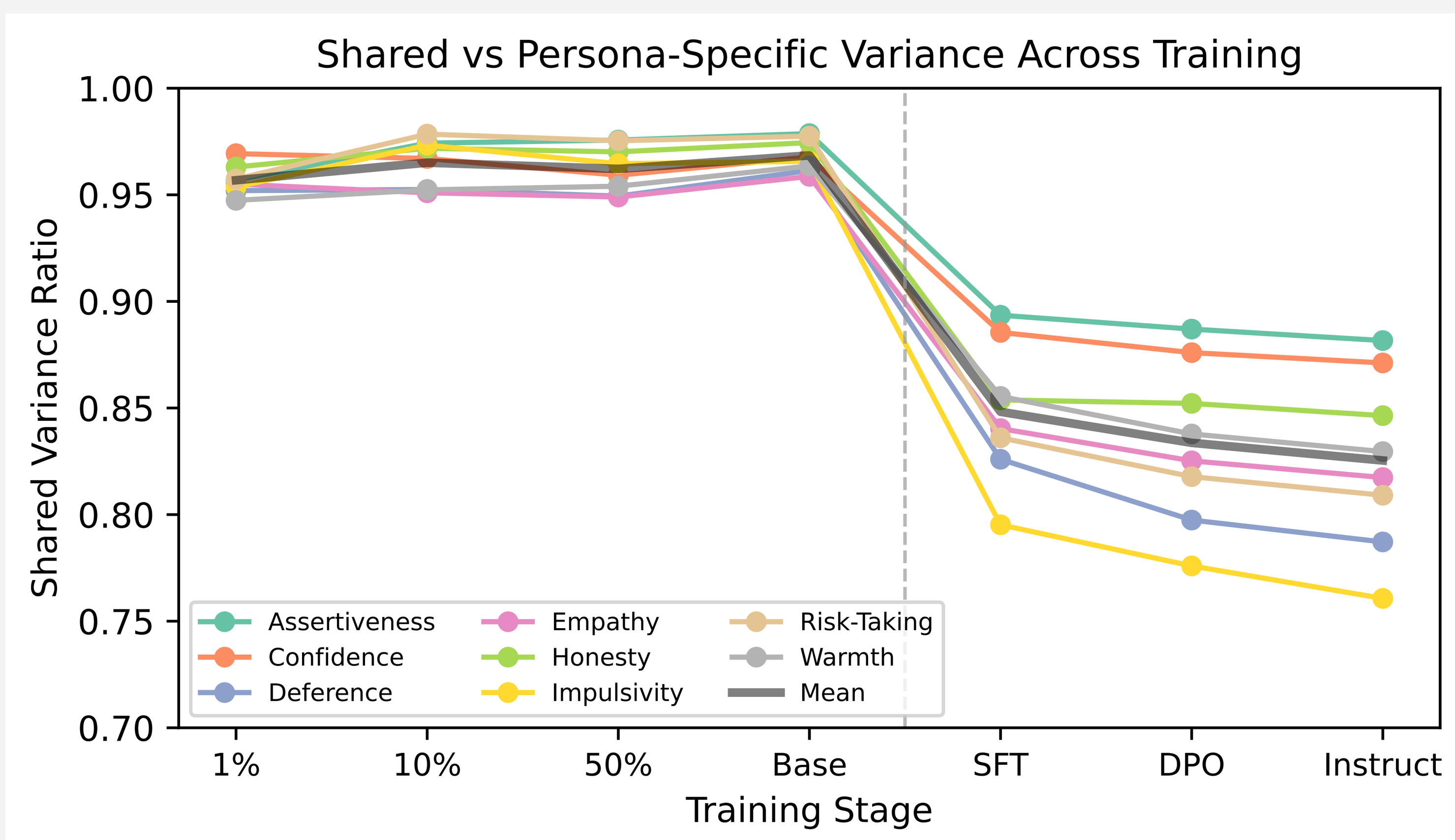


Shared vs Persona-Specific Variance



No trait exceeds 80% shared variance—all encode substantial persona-specific signal. Deference and impulsivity are most persona-conditioned (∼50%).

Training Trajectory: OLMo-2 7B



Shared variance ratio across training stages (OLMo-2 7B). Persona-specific structure emerges at SFT/DPO—pretraining checkpoints show near-uniform vectors regardless of persona.

Key Observations

- **Same trait, different vector**—persona identity reshapes encoding.
- **RLHF collapses diversity:** honesty shows high cross-persona similarity.
- **Intuitive pairs align:** K. teacher ≈ therapist on empathy.
- **Source "residues":** cross-steering leaks source biases into target.

Implications for Safety

Safety work assumes vectors are **model-global**—one sycophancy vector works everywhere.

Our results challenge this: **persona-specific representations** mean a single intervention may not generalise across deployment contexts.

Theory of change: Safety teams may need **persona-aware interventions** rather than one-size-fits-all vectors.

Next Steps & References

- Track across character training
- Safety traits: sycophancy, refusal
- Trait coupling (side-effects)
- More models (Llama, pre-instruct models)

Chen et al. (2025). *Persona Vectors.* arXiv:2507.21509.

Lu et al. (2026). *The Assistant Axis.* arXiv:2601.10387.

[1] We also extracted vectors using an *instruction-variant* method (opposing system-prompt instructions over shared questions). The key finding unique to that method: honesty converges to high shared variance across personas, consistent with RLHF collapsing representational diversity for heavily optimised traits.